Finding the best cell and aligning the antenna

ZTE MC889 5G CPE · with the ZTE helper script (zte-script-legacy.js) and the ZTE Helper Mac app

This guide walks through two jobs: **A** – physically positioning and aiming the router for the strongest, cleanest signal, and **B** – finding and locking the best cell. Do A first: in practice the position of the router changes the signal far more than the choice of cell.

Written for firmware YETTEL_BG_MC889V2.0.0B11 on a 5G NSA-only network (Yettel BG). The example numbers are readings taken from this router while the tooling was built.

> Everything the tooling shows is a value the router reports, or a simple average of those values. Nothing is inferred: if the router does not report something for a cell, the cell's field stays empty.

1. Before you start

- **Load the tooling.** Either open the router page in Firefox and paste the script into the developer console, or start the ZTE Helper app (*python3 zte_app.py* or *ZTE Helper.app*), which injects it automatically. Use **http://192.168.6.3** – the router's https certificate is self-signed.
- **Log in.** Use *Enable Automatic Login* once so reloads and reboots log you back in.
- **Do not use the "5G SA" network-mode link.** Your operator supports 5G NSA only; SA mode disconnects you.
- **Cell locks need a reboot** and survive further reboots until removed. The **LOCKS** row always shows what is active.
- Saved positions, test results and learned cell ids are kept in the browser's local storage – Firefox and the Mac app keep separate copies.

2. What the numbers mean

Higher is better for every value below (−80 dBm is stronger than −95 dBm). The ranges are common rules of thumb, not operator specifications.

Metric	What it measures	Excellent	Good	Fair	Poor
RSRP (dBm)	Strength of the cell's reference signal.	≥ −80	−80 to −90	−90 to −100	< −100
RSRQ (dB)	Signal quality including load and interference.	≥ −10	−10 to −15	−15 to −20	< −20
SINR (dB)	Signal vs. interference and noise – the best predictor of speed.	≥ 20	13 to 20	0 to 13	< 0
RSSI (dBm)	Total received power in the channel (signal + interference + noise).	–	–	–	–

What to optimise for. In 5G NSA the LTE cell is the anchor (control and fallback), while the **n78 5G carrier (3.5 GHz, 100 MHz)** carries most of the data. Align for **5G SINR first**, then check that LTE stays solid. n78 is far more sensitive to walls, glass and direction than the low LTE bands.

> **Example from this router:** the serving LTE cell (B7, PCI 340) is only ~156 m away (timing advance 2), yet LTE RSRP is about −91 dBm with SINR around 4–9 dB, and the two 5G receive branches differ by ~8 dB (rx0 −91, rx1 −83). A weak signal from a tower that close points to obstruction or orientation – exactly what antenna alignment fixes.

3. The screen at a glance

3.1 Neighbour cell table (NGBR)

RAT	BAND ▲	ARFCN	PCI	RSRP	AVG	MIN/MAX	RSRQ	SINR	RSSI	DIST	CELL	NOTE	LOCK
LTE	B1	100	340	-101.1	-101.1	-101.1/-101.1	-19.6	8.8	-72.5			**CA**	lock
LTE	B3	1550	434	-87	-87.0	-87/-87	-13		-65			**BETTER +4.2**	lock
LTE	B3	1550	340	-91.2	-91.2	-91.2/-91.2	-15.9	8.3	-66.3			**CA**	lock
LTE	B7	3350	340	-91	-89.0	-91/-87	-14	4.6	-55	~156 m	318269	**SERVING**	lock
LTE	B20	6400	112	-76	-76.0	-76/-76	-10		-56			**BETTER +6.4**	lock
LTE	B20	6400	394	-81	-81.0	-81/-81	-15		-56				lock
LTE	B20	6400	121	-82.4	-82.4	-82.4/-82.4	-15.8	6.3	-56.7			**CA**	lock
NR	n78	636576	127	-84	-84.0	-84/-84	-12	10.3	-74.0			**SERVING**	lock

Neighbour table with test data. Click any header to sort; click again to reverse.

Column	Meaning
RAT / BAND / ARFCN / PCI	Technology, band (derived from the frequency number), frequency number and physical cell id. A PCI is only unique within its frequency – the same PCI on two bands says nothing about being the same tower.
RSRP	Latest reading.
AVG, MIN/MAX	RSRP mean and range over the last 60 s (hover MIN/MAX for the sample count). Sort by MIN/MAX for the most stable cells.
RSRQ, SINR, RSSI	SINR exists only for cells you are connected to (SERVING / CA) – the router does not report it for neighbours.
DIST	Distance from the timing advance, shown only for the LTE serving cell (1 TA step = 78 m, ±39 m).
CELL	Cell identifier (e.g. 318269 – the value CellMapper searches by), learned while the router was connected to that exact cell (frequency + PCI); hover for eNB and sector (318269 = eNB 1243, sector 61). Grows as you try locks.
NOTE	SERVING, CA (carrier-aggregation cell) or **BETTER +x**: a neighbour whose 60 s mean RSRP beats the connected cell on the same frequency by ≥ 3 dB. Greyed rows were not reported in the last few seconds.
LOCK	One-click cell lock for that row (confirmation, then reboot).

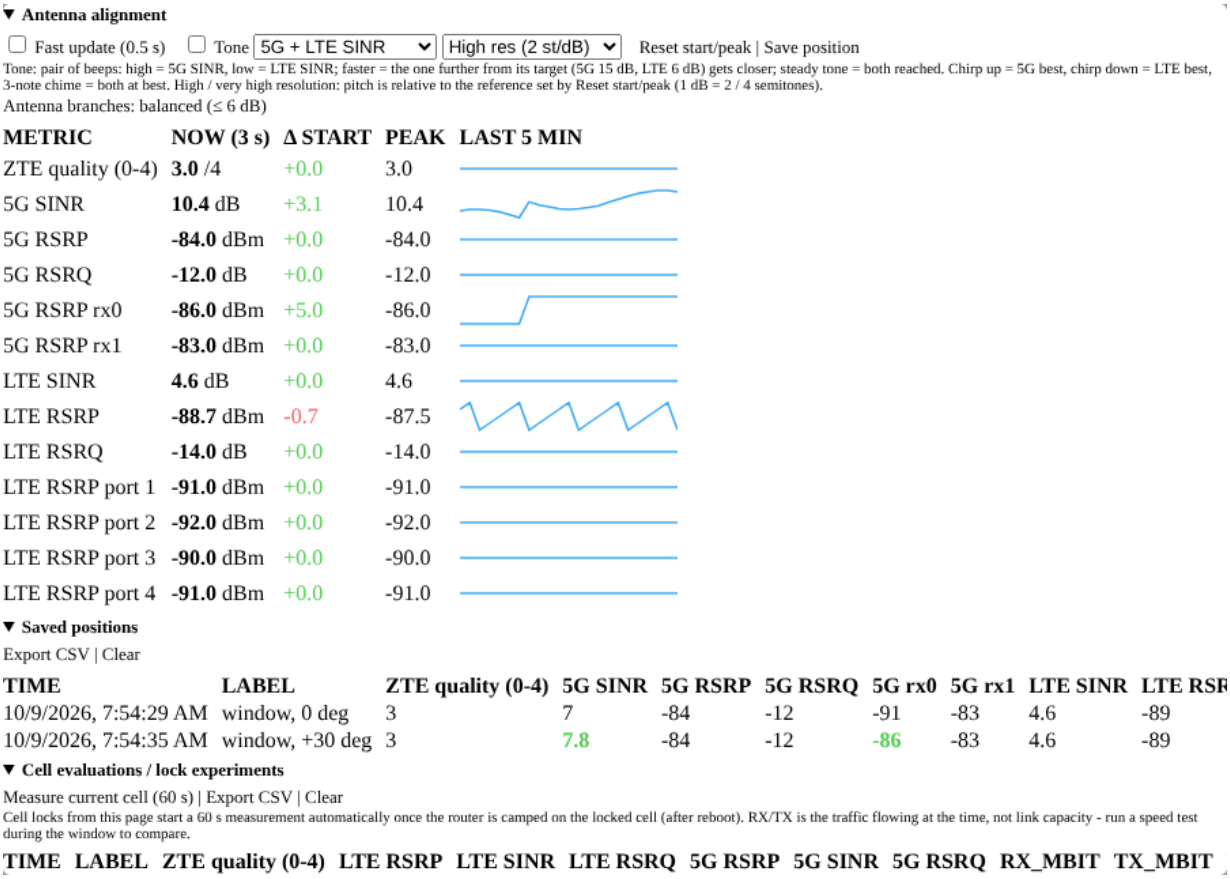
3.2 Status rows

LOCKS:	LTE cell: none 5G cell: none LTE bands: B1 B3 B7 B8 B20 B28 B38 B41 B43 5G NSA bands: n1 n3 n7 n8 n20 n28 n78

LOCKS row: active cell locks (orange when set) and the allowed LTE / 5G bands.

ANT. BRANCHES warns when the router's receive branches disagree by more than 6 dB (5G rx0/rx1, LTE ports 1–4). A large difference usually means the router's orientation or polarisation does not suit the signal – rotate it.

3.3 Alignment and test panel



▼ Antenna alignment

☐ Fast update (0.5 s) ☐ Tone 5G + LTE SINR High res (2 st/dB) Reset start/peak | Save position

Tone: pair of beeps: high = 5G SINR, low = LTE SINR; faster = the one further from its target (5G 15 dB, LTE 6 dB) gets closer; steady tone = both reached. Chirp up = 5G best, chirp down = LTE best, 3-note chime = both at best. High / very high resolution: pitch is relative to the reference set by Reset start/peak (1 dB = 2 / 4 semitones).

Antenna branches: balanced (≤ 6 dB)

METRIC	NOW (3 s)	Δ START	PEAK	LAST 5 MIN
ZTE quality (0-4)	**3.0** /4	+0.0	3.0	
5G SINR	**10.4** dB	+3.1	10.4	
5G RSRP	**-84.0** dBm	+0.0	-84.0	
5G RSRQ	**-12.0** dB	+0.0	-12.0	
5G RSRP rx0	**-86.0** dBm	+5.0	-86.0	
5G RSRP rx1	**-83.0** dBm	+0.0	-83.0	
LTE SINR	**4.6** dB	+0.0	4.6	
LTE RSRP	**-88.7** dBm	-0.7	-87.5	
LTE RSRQ	**-14.0** dB	+0.0	-14.0	
LTE RSRP port 1	**-91.0** dBm	+0.0	-91.0	
LTE RSRP port 2	**-92.0** dBm	+0.0	-92.0	
LTE RSRP port 3	**-90.0** dBm	+0.0	-90.0	
LTE RSRP port 4	**-91.0** dBm	+0.0	-91.0	

▼ Saved positions

Export CSV | Clear

TIME	LABEL	ZTE quality (0-4)	5G SINR	5G RSRP	5G RSRQ	5G rx0	5G rx1	LTE SINR	LTE RSF
10/9/2026, 7:54:29 AM	window, 0 deg	3	7	-84	-12	-91	-83	4.6	-89
10/9/2026, 7:54:35 AM	window, +30 deg	3	7.8	-84	-12	-86	-83	4.6	-89

▼ Cell evaluations / lock experiments

Measure current cell (60 s) | Export CSV | Clear

Cell locks from this page start a 60 s measurement automatically once the router is camped on the locked cell (after reboot). RX/TX is the traffic flowing at the time, not link capacity - run a speed test during the window to compare.

TIME	LABEL	ZTE quality (0-4)	LTE RSRP	LTE SINR	LTE RSRQ	5G RSRP	5G SINR	5G RSRQ	RX_MBIT	TX_MBIT

Antenna alignment panel with test data: 3 s averages, change since start, peak and a 5-minute graph per value; saved positions with the best value per column in green.

- **Fast update (0.5 s)** – polls twice per second while you move the router.
- **Tone** (off by default) – sounds on the computer speakers. The drop-down picks the mode (a legend under the controls explains the current one):
- **Reset start/peak** – start a fresh sweep; Δ START, PEAK and the tone's "best" restart from now.
- **Save position** – stores 5 s averages under a label you type.
- **Measure current cell (60 s)** – records 60 s averages and traffic for the cell you are on.

Tone mode	What you hear
5G + LTE SINR (default)	A pair of beeps: **high note = 5G SINR**, **low note = LTE SINR**; each pitch follows its SINR, and the beeps speed up as the one further from its target gets closer – targets **5G 15 dB, LTE 6 dB** (LTE SINR is typically several dB lower here). **Steady tone** = both targets reached. **Rising chirp** = new 5G best, **falling chirp** = new LTE best, **three-note rising chime** = 5G and LTE both at their best at the same time. Use this for the main alignment: download runs on 5G, upload and control on LTE.
5G rx0/rx1 balance	Target: rx0 − rx1 = 0. **Steady tone** = the two 5G receive branches are within the balanced window (2 / 1 / 0.5 dB, see Resolution). Otherwise beeps: **low = rx0 stronger**, **high = rx1 stronger**; the beeps get faster as the difference shrinks, then merge into the steady tone.
ZTE quality (0-4)	The router's own signal grade (0 Poor … 4 Excellent+) – the value ZTE's installer app aligns with. One pitch step and faster beeps per level, **steady tone at 4**, chirp on every level up. The simplest way to align; use the SINR modes for the fine detail within a level.

Tone mode	What you hear
ZTE quality + 5G SINR	One aggregated tone: **ZTE quality sets the pitch step** (one step per level), **5G SINR moves it slightly within the step** (never into the next level) and sets the beep speed (faster = closer to 15 dB). **3-note chime** = quality level up, **rising chirp** = new 5G SINR best, **steady tone** = quality 4 and 5G SINR ≥ 15 dB. Coarse and fine alignment in one sound.
Single value (e.g. 5G SINR)	Faster beeps the closer the value is to its target – **5G SINR 20 dB, LTE SINR 6 dB, RSRQ −10 dB, RSRP −70 dBm** (editable in TONE_TARGET at the top of the tone code); **steady tone** = target reached; double chirp on a new best.

Resolution (second drop-down) – for weak signals where changes are small:

Resolution	Pitch	Balanced window	New-best step
Absolute	Fixed scale from a floor up to the target (5G SINR −10 to 20 dB, LTE SINR −10 to 6 dB, RSRQ −20 to −10 dB, RSRP −120 to −70 dBm); small changes are hard to hear	2 dB	0.5 dB
High res (default)	Relative to the reference: **1 dB = 2 semitones**	1 dB	0.3 dB
Very high	Relative to the reference: **1 dB = 4 semitones**	0.5 dB	0.2 dB

With high / very high resolution the pitch starts in the middle and moves up or down from the **reference** – the value when the tone was switched on or **Reset start/peak** was clicked. Click Reset at the start of each sweep to re-centre it. Beep speed always shows how close you are to the target; the resolution only changes the pitch. The tone uses a 1.5 s average (use **Fast update**) so it reacts quickly; the table keeps the steadier 3 s average. At most one chirp per second.

4. Part A – aligning the antenna precisely

Goal: the highest stable **5G SINR**, with balanced receive branches and good LTE. Work in this order: location, then direction, then fine tuning.

Step 1 – choose the location

- Put the router at a window or outer wall facing the serving tower, as high as practical, away from metal, mirrors and other electronics.
- Modern coated (low-E) glass can block 3.5 GHz badly – compare “window closed” and “window open” once to see how much your glass costs.
- Use **DIST** as a sanity check: a close tower with weak RSRP means something is in the way.

Step 2 – prepare the panel

1. Open **Antenna alignment**, tick **Fast update (0.5 s)**, click **Reset start/peak**.
2. For a quick first pass use the **ZTE quality (0-4)** tone, or **ZTE quality + 5G SINR** to hear the level and the fine 5G detail together. For precise work switch the **Tone** to **5G + LTE SINR** mode with **High res** (or **Very high** for a weak signal) if you want to work without looking at the screen.
3. Step back from the router while it measures – your body near the router lowers the reading.

Step 3 – coarse sweep

1. Rotate the router in 30–45° steps through a full turn (or along the window).
2. At each step **wait at least 5 s** (the values are 3 s averages) and click **Save position** with a label such as “+45°”.
3. Open **Saved positions**: the best value per column is green. Pick the position that wins on 5G SINR without hurting LTE much.

Step 4 – fine tuning

1. Return to the best position, click **Reset start/peak**, then turn in 10–15° steps either side.
2. Listen for the chirps (rising = 5G best, falling = LTE best) and aim for the **three-note chime**, which means both are at their best together; the **steady tone** means 5G reached 15 dB and LTE 6 dB. Turn past the best point until the values clearly drop, then come back.
3. Try small tilts and a few centimetres of shift – at 3.5 GHz a short move can change SINR noticeably.
4. Balance the 5G branches: switch the tone to **5G rx0/rx1 balance** and rotate the router around its vertical axis in small steps until you hear the **steady tone**; low beeps = rx0 stronger, high beeps = rx1 stronger. **ANT. BRANCHES** shows the same on screen. Then switch back to 5G + LTE SINR to confirm the SINRs did not drop.
5. Save the final position.

Step 5 – confirm

1. Run **Measure current cell (60 s)** while running a speed test, and keep the result.
2. Check again at a busy time of day: load changes RSRQ and SINR, so a position that only looks good at night is not the winner.
3. Export the saved positions to CSV if you want to keep the record.

Trust the averages and the graph, not single readings – router values jump by a few dB between updates. A position is better only if the difference stays visible for several seconds.

5. Part B – finding the best cell

Only start once the antenna is aligned – a new position changes which cells are strongest.

Step 1 – observe

1. Let the page run for 1–2 minutes so every row has 60 s statistics (the router reports different neighbours over time).
2. Sort by **AVG** (strongest first) and by **MIN/MAX** (most stable first). A cell that is steady at −80 dBm beats one jumping between −75 and −90.
3. Note any **BETTER +x** rows. Example from this router: B20 PCI 112 at about −76 dBm vs. the B20 CA cell PCI 121 at −82 dBm – a candidate worth testing.

Step 2 – baseline

Click **Measure current cell (60 s)** and run a speed test inside the minute. This is the result every lock has to beat.

Step 3 – test a candidate

1. Click **lock** on the candidate row and confirm. LTE locks use PCI + EARFCN; 5G locks use PCI, ARFCN, band and SCS (30 kHz for n78, 15 kHz for low bands; for ambiguous bands you are asked to complete the values).
2. Accept the reboot. After the router is back, log in (automatic if enabled).
3. The test starts by itself as soon as the router is connected to the locked cell, measures for 60 s and stores the result under **Cell evaluations / lock experiments**. Run a speed test during that minute – RX/TX shows traffic, not capacity.
4. If the router does not connect to that cell within 10 minutes, or leaves it during the test, that is recorded as the result.

Step 4 – decide

1. Compare the results table against the baseline: 5G SINR and speed first, then LTE SINR/RSRQ.
2. Keep the best lock, or use **Remove LTE Cell Lock** / **Remove 5G Cell Lock** (and reboot) to return to automatic selection.
3. Check the **LOCKS** row afterwards so no forgotten lock remains.

NSA caution: the 5G leg is added on top of the LTE anchor. Locking LTE to a cell whose site has no n78 can cost you 5G entirely – after every lock check that the 5G (n78) row is still present.

Lighter alternative: band locks

LTE Bands and **5G Bands** restrict the router to chosen bands without fixing a single cell. They are quicker to try and leave the router free to pick the best cell within those bands. Use *AUTO* to undo.

6. Quick checklist

#	Action	Tooling
1	Router at window / outer wall facing the tower	DIST, 5G SINR
2	Fast update on, Reset start/peak, tone optional (5G + LTE SINR mode)	Antenna alignment
3	Coarse sweep in 30–45° steps, 5 s per step, Save position each	Saved positions
4	Fine tune ±10–15°, tilt, small shifts; aim for the 3-note chime	PEAK, tone (5G + LTE SINR)
5	Balance 5G rx0/rx1 until the steady tone	tone (balance), ANT. BRANCHES
6	Baseline 60 s measurement with a speed test	Measure current cell
7	Observe neighbours 1–2 min; sort by AVG and MIN/MAX; note BETTER hints	NGBR table
8	Lock a candidate, reboot, let the 60 s test run with a speed test	lock, lock experiments
9	Keep the winner or remove the lock; confirm 5G is still present	LOCKS row, 5G row

7. Troubleshooting

Problem	What to do
No sound with Tone on	Untick and tick Tone again (browsers only start audio after a click) and check the Mac's output volume.
A value shows “?”	The router sent its “no value” marker (for example SINR −20.0 or −3276.8).
Router has no service after a lock	The locked cell is not reachable. The web page still works over the LAN: use Remove LTE / 5G Cell Lock and reboot.
Lock experiment never starts	The router is not connected to the locked cell yet, or the lock was not applied because the router was not rebooted.
Export CSV does nothing in the Mac app	Use Firefox for exports; the data is in the browser where you saved it.
SINR missing for a neighbour	Expected – the router only reports SINR for connected cells.
Signal Quality Test / buzzer	Not available: this firmware accepts the start command but never reports results, and rejects the buzzer command.

Router fields used: Z5g_rsrp, Z5g_rsrq, Z5g_SINR, Z5g_rssi, 5g_rx0_rsrp, 5g_rx1_rsrp, lte_rsrp, lte_rsrq, lte_snr, lte_rssi, lte_rsrp_1–4, lte_ta, cell_id, lte_pci, wan_active_channel, lte_multi_ca_scell_info, lte_multi_ca_scell_sig_info, nr_multi_ca_scell_info, ngbr_cell_info, lte_pci_lock, lte_earfcn_lock, nr5g_cell_lock, lte_band_lock, nr5g_nsa_band_lock, realtime_rx_thrpt, realtime_tx_thrpt.

8. Maintaining the tooling: tests

The script has automated checks that run on any computer with Node.js 18+ (no installation of packages) – no router needed. Run them after every edit of *zte-script-legacy.js*, before pasting it into Firefox or restarting the Mac app. Details are also in *tests/README.md* next to the test file.

Run

```
cd legacy
node --check zte-script-legacy.js                # syntax
node tests/test_cmp.js zte-script-legacy.js      # logic → ALL PASSED
python3 app/zte_app.py --selftest                # Mac app prompt bridge
```

Each case prints **ok** or **FAIL** with the actual values; the run ends with ALL PASSED or the number of failures and exits with code 1 on failure, so it can be used as a git pre-commit hook:

```
node --check legacy/zte-script-legacy.js && node legacy/tests/test_cmp.js
legacy/zte-script-legacy.js >/dev/null
```

What is covered

Area	Checks
Neighbour table sorting	Both directions with negative/negative, negative/positive and positive/positive values; 0 is a real value; missing values always last; tie-breaks.
Neighbour table content	BETTER hint, MIN/MAX, connected cells (no NaN PCI, stale 5G CA cells removed), learned cell id survives cell ping-pong.
Alignment table	Change since start (sign, colour, no red −0.0), peak, no freeze after Reset with an invalid sample.
Saved positions / results	Best value per column; labels are HTML-escaped.
Tones	Targets (RSRP −70 dBm, RSRQ −10 dB, 5G SINR 15/20 dB, LTE SINR 6 dB), closeness direction, relative pitch, rx0/rx1 direction, branch warning.
Lock experiment	Waits through the reboot delay, waits again if the target cell is lost, stores a 60 s result.

How the test works

- It reads the script as text. **grab('name')** cuts a named function out and evaluates it in Node; **vars(regex)** loads the global tables (NGBR_COLUMNS, LTE_BANDS / NR_BANDS, ALIGN_METRICS, TONE_TARGET, TONE_RES, EXP_COLS, EXP_FIELDS).
- jQuery **$**, **tools_store / tools_save** and **localStorage** are small stubs; whatever the code writes with $(sel).html(v) is captured for the assertions.
- Router values are set in one fixture, as strings exactly like the router sends them: **lte_pci, nr5g_pci and cell_id are hex**, PCIs inside ngbr_cell_info and lte_multi_ca_scell_info are **decimal**.

Keeping the tests up to date

When you…	Do this
rename a function, or a test needs a new one	Update the function list in *for (const f of [...])* near the top of test_cmp.js.
change a global table	Keep it one *var NAME = ...;* statement, or update the matching vars(...) regex.
read a new router field in tested code	Add it to the fixture (a missing one fails with ReferenceError).
change a target or threshold (TONE_TARGET, TONE_RES, BETTER 3 dB, branch 6 dB)	Update the expected values of the affected cases – the failure is a reminder that behaviour changed.

When you…	Do this
add a comparison or calculation	Add *t('what it proves', () => { … assert … })* covering negative/negative, negative/positive, positive/positive, zero and missing values.
see a failing case	Read the actual values in the message; fix the script, not the expectation, unless the change was intended.